

4.11.38

EE25BTECH11061 - V.Sainadh

Question: Find the area of the region bounded by the lines $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.

Solution: Given three lines are

$$\begin{pmatrix} -2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 1 \implies \mathbf{n}^\top \mathbf{x} = 1 \quad (1)$$

$$\begin{pmatrix} -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 1 \implies \mathbf{m}^\top \mathbf{x} = 1 \quad (2)$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 4 \implies \mathbf{p}^\top \mathbf{x} = 4 \quad (3)$$

The three lines form a triangle. The vertices of triangle are obtained by

Intersection of : (1) and (3)

$$\left(\begin{array}{cc|c} -2 & 1 & 1 \\ 1 & 0 & 4 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + 2R_2} \left(\begin{array}{cc|c} 0 & 1 & 9 \\ 1 & 0 & 4 \end{array} \right) \quad (1)$$

From the second row we get $y = 9$ so $x = 4 \implies \mathbf{A} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$

Intersection of : (2) and (3)

$$\left(\begin{array}{cc|c} -3 & 1 & 1 \\ 1 & 0 & 4 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + 3R_2} \left(\begin{array}{cc|c} 0 & 1 & 13 \\ 1 & 0 & 4 \end{array} \right) \quad (2)$$

From the second row we get $y = 13$ so $x = 4 \implies \mathbf{B} = \begin{pmatrix} 4 \\ 13 \end{pmatrix}$

Intersection of : (1) and (2)

$$\left(\begin{array}{cc|c} -2 & 1 & 1 \\ -3 & 1 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{cc|c} -2 & 1 & 1 \\ -1 & 0 & 0 \end{array} \right) \quad (3)$$

From the second row we get $x = 0$ so $y = 1 \implies \mathbf{C} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}, \quad \mathbf{A} - \mathbf{C} = \begin{pmatrix} 4 \\ 8 \end{pmatrix} \quad (4)$$

$$\|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| = \begin{vmatrix} 0 & -4 \\ 4 & 8 \end{vmatrix} = |0 - (-16)| = 16 \quad (5)$$

$$\text{Area of the triangle} = \frac{1}{2} \|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| = \frac{16}{2} = 8 \quad (6)$$

Area of the triangle formed by the lines $y = 2x + 1$, $y = 3x + 1$ and $x = 4$ is 8 square units.

Graph:

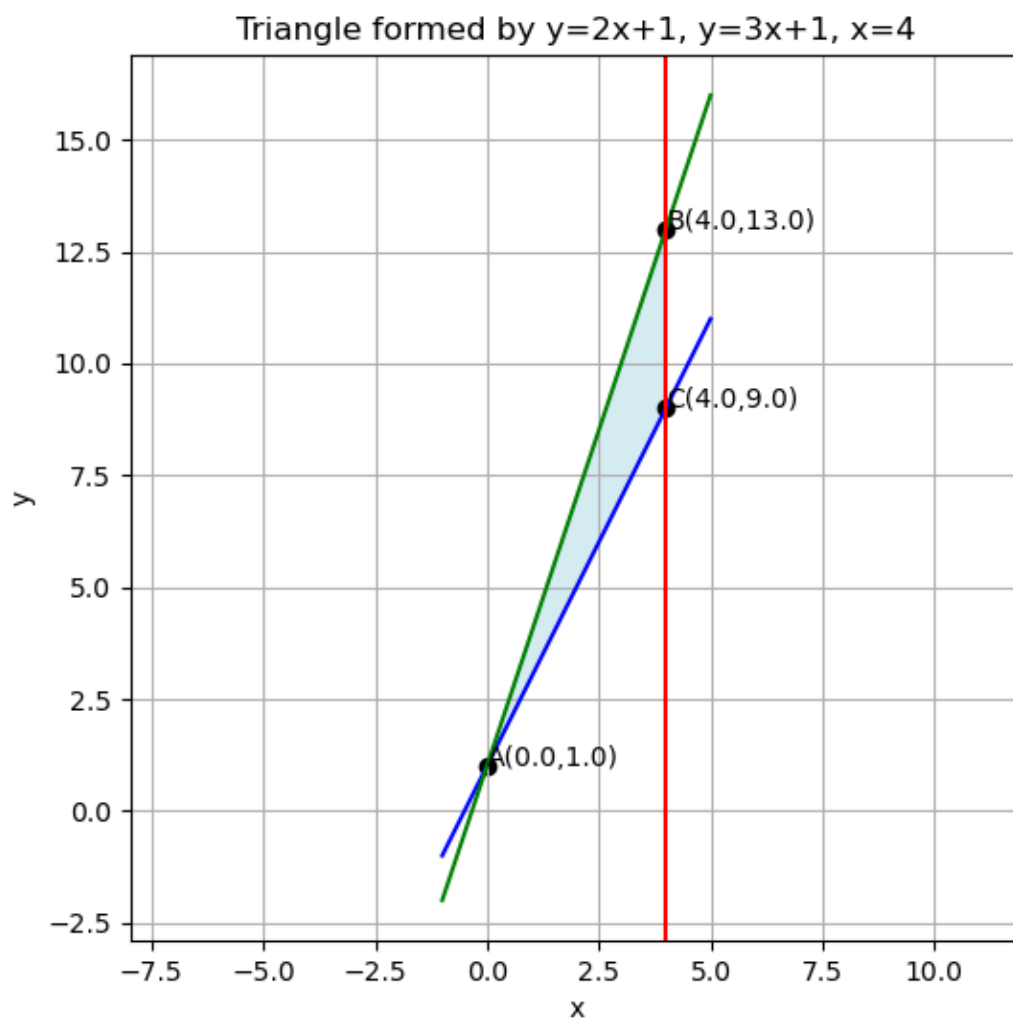


Fig. 0: Representation of a Triangle